



BLACK SMOKERS

The mid-ocean ridges on deep ocean floors are dotted with hot springs that gush superheated water into the cold, dark ocean. The hot water contains dissolved minerals that solidify into tiny sooty particles as soon as they hit the near-freezing ocean, so the erupting water can look like a cloud of billowing black smoke. The minerals often build up to form dramatic “chimneys” that tower above the vents.

▲ DARK MATTER

Lit up by the powerful lights of a deepwater submersible vehicle, a black smoker belches hot water and mineral particles into the cold ocean.



HOT BATH

The hot springs of Nagano, Japan, feed a series of pools where the water cools to a comfortable 122°F (50°C) or so. Japanese macaques, or snow monkeys, use them to keep warm in winter, when the air temperature in the region can drop to a numbing 5°F (–15°C). The macaques sit in the steaming pools, basking in the volcanic heat while snow falls on their heads.



PRESSURE COOKER

At a few sites, hot water beneath the ground erupts as geysers. The weight of the water above the heat source increases the pressure on the hot water, and this raises its boiling point. It gets hotter and hotter, and expands until some of the water is pushed up and out of the ground. This reduces the pressure, so the superheated water boils and turns to steam—blowing the remaining water out of the ground as a geyser.

◀ BLOWOUT

A geyser in the Taupo volcanic zone of New Zealand shoots steam and hot water high into the air. Some geysers erupt to heights of well over 330 ft (100 m).